Judson, *Abstract Algebra: Theory and Applications*
The Structure of Groups: Solutions to Exercises 1–8, 10, 11, 17, 18, 22–24

Exercises 9, 12–16 (solvable groups) and 19–21 are not included. I did not study these topics yet.

Facts used below

Fundamental Theorem of Finitely Generated Abelian Groups. Every finitely generated abelian group is isomorphic to $\mathbb{Z}^r \times \mathbb{Z}_{p_1^{e_1}} \times \cdots \times \mathbb{Z}_{p_k^{e_k}}$ (the p_i not necessarily distinct primes), and r and the multiset of prime powers $p_i^{e_i}$ are uniquely determined. Equivalently, a finite abelian group is $\mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_t}$ with $n_1 \mid n_2 \mid \cdots \mid n_t$ (the *invariant factors*), uniquely determined. A finite abelian group of order $p_1^{a_1} \cdots p_s^{a_s}$ is determined by choosing, for each prime p_i, a partition of a_i.

Fact A (lifting a series from G/N). Let $N \trianglelefteq G$ and $\pi : G \to G/N$ the canonical map. Let $\bar{K} \trianglelefteq \bar{L}$ be subgroups of G/N, and $K = \pi^{-1}(\bar{K})$, $L = \pi^{-1}(\bar{L})$. Then $N \subseteq K \subseteq L$, $K \trianglelefteq L$, and $L/K \cong \bar{L}/\bar{K}$.

Proof. The composite $L \xrightarrow{\pi} \bar{L} \to \bar{L}/\bar{K}$ is a surjective homomorphism, with kernel $\{x \in L : \pi(x) \in \bar{K}\} = K$. Apply the First Isomorphism Theorem. □

Exercise 1

Find all of the abelian groups of order less than or equal to 40 up to isomorphism.

Solution

By the Fundamental Theorem, the number of abelian groups of order $n = \prod p_i^{a_i}$ is $\prod P(a_i)$, where $P(a)$ is the number of partitions of a. If n is squarefree, the only abelian group of order n is $\mathbb{Z}_n$. This covers

$$n = 1, 2, 3, 5, 6, 7, 10, 11, 13, 14, 15, 17, 19, 21, 22, 23, 26, 29, 30, 31, 33, 34, 35, 37, 38, 39.$$

For the remaining orders:

n	abelian groups of order n
4	$\mathbb{Z}_4$; $\mathbb{Z}_2 \times \mathbb{Z}_2$
8	$\mathbb{Z}_8$; $\mathbb{Z}_4 \times \mathbb{Z}_2$; $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$
9	$\mathbb{Z}_9$; $\mathbb{Z}_3 \times \mathbb{Z}_3$
12	$\mathbb{Z}_{12}$; $\mathbb{Z}_2 \times \mathbb{Z}_6$
16	$\mathbb{Z}_{16}$; $\mathbb{Z}_8 \times \mathbb{Z}_2$; $\mathbb{Z}_4 \times \mathbb{Z}_4$; $\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2$; $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$
18	$\mathbb{Z}_{18}$; $\mathbb{Z}_3 \times \mathbb{Z}_6$
20	$\mathbb{Z}_{20}$; $\mathbb{Z}_2 \times \mathbb{Z}_{10}$
24	$\mathbb{Z}_{24}$; $\mathbb{Z}_2 \times \mathbb{Z}_{12}$; $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6$
25	$\mathbb{Z}_{25}$; $\mathbb{Z}_5 \times \mathbb{Z}_5$
27	$\mathbb{Z}_{27}$; $\mathbb{Z}_9 \times \mathbb{Z}_3$; $\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3$
28	$\mathbb{Z}_{28}$; $\mathbb{Z}_2 \times \mathbb{Z}_{14}$
32	$\mathbb{Z}_{32}$; $\mathbb{Z}_{16} \times \mathbb{Z}_2$; $\mathbb{Z}_8 \times \mathbb{Z}_4$; $\mathbb{Z}_8 \times \mathbb{Z}_2 \times \mathbb{Z}_2$; $\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_2$; $\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$; $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$
36	$\mathbb{Z}_{36}$; $\mathbb{Z}_2 \times \mathbb{Z}_{18}$; $\mathbb{Z}_3 \times \mathbb{Z}_{12}$; $\mathbb{Z}_6 \times \mathbb{Z}_6$
40	$\mathbb{Z}_{40}$; $\mathbb{Z}_2 \times \mathbb{Z}_{20}$; $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{10}$

(All groups are written in invariant factor form, e.g. $\mathbb{Z}_2 \times \mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3$.) This gives $26+42 = 68$ abelian groups in total.

Exercise 2

Find all of the abelian groups of order 200 up to isomorphism.

Solution

$200 = 2^3 \cdot 5^2$. There are $P(3)P(2) = 3 \cdot 2 = 6$ groups (partitions of 3: $3, 2+1, 1+1+1$; of 2: $2, 1+1$):

primary decomposition	invariant factors
$\mathbb{Z}_8 \times \mathbb{Z}_{25}$	$\mathbb{Z}_{200}$
$\mathbb{Z}_8 \times \mathbb{Z}_5 \times \mathbb{Z}_5$	$\mathbb{Z}_5 \times \mathbb{Z}_{40}$
$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_{25}$	$\mathbb{Z}_2 \times \mathbb{Z}_{100}$
$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_5$	$\mathbb{Z}_{10} \times \mathbb{Z}_{20}$
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{25}$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{50}$
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_5$	$\mathbb{Z}_2 \times \mathbb{Z}_{10} \times \mathbb{Z}_{10}$

Exercise 3

Find all of the abelian groups of order 720 up to isomorphism.

Solution

$720 = 2^4 \cdot 3^2 \cdot 5$. There are $P(4)P(2)P(1) = 5 \cdot 2 \cdot 1 = 10$ groups.

	primary decomposition	invariant factors
1	$\mathbb{Z}_{16} \times \mathbb{Z}_9 \times \mathbb{Z}_5$	$\mathbb{Z}_{720}$
2	$\mathbb{Z}_{16} \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$	$\mathbb{Z}_3 \times \mathbb{Z}_{240}$
3	$\mathbb{Z}_8 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$	$\mathbb{Z}_2 \times \mathbb{Z}_{360}$
4	$\mathbb{Z}_8 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$	$\mathbb{Z}_6 \times \mathbb{Z}_{120}$
5	$\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_5$	$\mathbb{Z}_4 \times \mathbb{Z}_{180}$
6	$\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$	$\mathbb{Z}_{12} \times \mathbb{Z}_{60}$
7	$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{180}$
8	$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$	$\mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{60}$
9	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{90}$
10	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{30}$

Exercise 4

Find all of the composition series for each of the following groups.

(a) $\mathbb{Z}_{12}$

(b) $\mathbb{Z}_{48}$

(c) The quaternions, Q_8

(d) D_4

(e) $S_3 \times \mathbb{Z}_4$

(f) S_4

(g) S_n, $n \geq 5$

(h) $\mathbb{Q}$

Solution

A composition series is a chain $\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_k = G$ in which each G_i is normal in G_{i+1} (not necessarily in G) and each factor G_{i+1}/G_i is simple. For the finite solvable groups below, the simple factors are exactly the groups of prime order. By the Jordan–Hölder Theorem, the factors of any two composition series of the same group agree up to order and isomorphism.

(a) $\mathbb{Z}_{12}$. The group is abelian, so every subgroup is normal and the composition series are the chains of subgroups in which each step has prime index. Subgroups: $\langle d \rangle$ for $d \mid 12$ (of order $12/d$). There are three composition series:

$$\{0\} \subset \langle 6 \rangle \subset \langle 2 \rangle \subset \mathbb{Z}_{12} \qquad \text{(factors } \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_2\text{)},$$
$$\{0\} \subset \langle 6 \rangle \subset \langle 3 \rangle \subset \mathbb{Z}_{12} \qquad \text{(factors } \mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3\text{)},$$
$$\{0\} \subset \langle 4 \rangle \subset \langle 2 \rangle \subset \mathbb{Z}_{12} \qquad \text{(factors } \mathbb{Z}_3, \mathbb{Z}_2, \mathbb{Z}_2\text{)}.$$

(b) $\mathbb{Z}_{48}$. Again every chain with prime index at each step works. Since $48 = 2^4 \cdot 3$, the factor $\mathbb{Z}_3$ can occur in any of the 5 positions, giving 5 composition series (write $\langle d \rangle$ for the subgroup of

order $48/d$):

$$\begin{aligned}
&\{0\} \subset \langle 24\rangle \subset \langle 12\rangle \subset \langle 6\rangle \subset \langle 3\rangle \subset \mathbb{Z}_{48} && (\mathbb{Z}_2,\mathbb{Z}_2,\mathbb{Z}_2,\mathbb{Z}_2,\mathbb{Z}_3),\\
&\{0\} \subset \langle 24\rangle \subset \langle 12\rangle \subset \langle 6\rangle \subset \langle 2\rangle \subset \mathbb{Z}_{48} && (\mathbb{Z}_2,\mathbb{Z}_2,\mathbb{Z}_2,\mathbb{Z}_3,\mathbb{Z}_2),\\
&\{0\} \subset \langle 24\rangle \subset \langle 12\rangle \subset \langle 4\rangle \subset \langle 2\rangle \subset \mathbb{Z}_{48} && (\mathbb{Z}_2,\mathbb{Z}_2,\mathbb{Z}_3,\mathbb{Z}_2,\mathbb{Z}_2),\\
&\{0\} \subset \langle 24\rangle \subset \langle 8\rangle \subset \langle 4\rangle \subset \langle 2\rangle \subset \mathbb{Z}_{48} && (\mathbb{Z}_2,\mathbb{Z}_3,\mathbb{Z}_2,\mathbb{Z}_2,\mathbb{Z}_2),\\
&\{0\} \subset \langle 16\rangle \subset \langle 8\rangle \subset \langle 4\rangle \subset \langle 2\rangle \subset \mathbb{Z}_{48} && (\mathbb{Z}_3,\mathbb{Z}_2,\mathbb{Z}_2,\mathbb{Z}_2,\mathbb{Z}_2).
\end{aligned}$$

(c) Q_8**.** The subgroups are $\{1\}$, $\{\pm 1\}$, $\langle i\rangle$, $\langle j\rangle$, $\langle k\rangle$, Q_8, all normal. The composition series are

$$\{1\} \subset \{\pm 1\} \subset \langle i\rangle \subset Q_8, \qquad \{1\} \subset \{\pm 1\} \subset \langle j\rangle \subset Q_8, \qquad \{1\} \subset \{\pm 1\} \subset \langle k\rangle \subset Q_8,$$

each with factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_2$. There are no others, since the only subgroups of order 2 and 4 are those listed.

(d) $D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1}\rangle$**.** Here $|D_4| = 8$, so a composition series has the form $\{e\} \subset A \subset B \subset D_4$ with $|A| = 2$, $|B| = 4$. Every subgroup B of order 4 has index 2, hence is normal in D_4, and $A \triangleleft B$ automatically since B is abelian. So we just need the pairs $A \subset B$. The subgroups of order 4 are

$$\langle r\rangle, \qquad V_1 = \{1, r^2, s, sr^2\}, \qquad V_2 = \{1, r^2, sr, sr^3\},$$

containing respectively the subgroups of order 2: $\langle r^2\rangle$; $\langle r^2\rangle, \langle s\rangle, \langle sr^2\rangle$; $\langle r^2\rangle, \langle sr\rangle, \langle sr^3\rangle$. Hence there are 7 composition series, all with factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_2$:

$$\begin{aligned}
&\{e\} \subset \langle r^2\rangle \subset \langle r\rangle \subset D_4,\\
&\{e\} \subset \langle r^2\rangle \subset V_1 \subset D_4, \quad \{e\} \subset \langle s\rangle \subset V_1 \subset D_4, \quad \{e\} \subset \langle sr^2\rangle \subset V_1 \subset D_4,\\
&\{e\} \subset \langle r^2\rangle \subset V_2 \subset D_4, \quad \{e\} \subset \langle sr\rangle \subset V_2 \subset D_4, \quad \{e\} \subset \langle sr^3\rangle \subset V_2 \subset D_4.
\end{aligned}$$

(e) $S_3 \times \mathbb{Z}_4$ (order 24). Write $G = S_3 \times \mathbb{Z}_4$ and define the following subgroups, where $A_3 = \langle (123)\rangle$:

$$\begin{aligned}
P &= A_3 \times \{0\} \text{ (order 3)}, & Z &= \{(1)\} \times \langle 2\rangle \text{ (order 2)},\\
C_4 &= \{(1)\} \times \mathbb{Z}_4 \text{ (order 4)}, & C &= A_3 \times \langle 2\rangle \cong \mathbb{Z}_6 \text{ (order 6)},\\
S &= S_3 \times \{0\} \text{ (order 6)}, & T &= \{(\sigma, z) : z = 0 \text{ if } \sigma \text{ even},\ z = 2 \text{ if } \sigma \text{ odd}\} \cong S_3.
\end{aligned}$$

The factors of any composition series are $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3$ in some order, so the orders of the terms are $1, \cdot, \cdot, \cdot, 24$ with each ratio prime.

The term G_3. It is normal of prime index in G. It has index 3 only if it is a normal Sylow 2-subgroup; but the Sylow 2-subgroups $\langle t\rangle \times \mathbb{Z}_4$ (t a transposition) are three distinct subgroups, so none is normal. Hence G_3 has index 2, and $\mathrm{Hom}(G, \mathbb{Z}_2) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ has exactly three non-trivial elements, giving three subgroups of index 2:

$$\begin{aligned}
&N_1 = A_3 \times \mathbb{Z}_4 \cong \mathbb{Z}_{12},\\
&N_2 = S_3 \times \langle 2\rangle \cong S_3 \times \mathbb{Z}_2,\\
&N_3 = \{(\sigma, z) : \sigma \text{ even},\ z \text{ even}\} \cup \{(\sigma, z) : \sigma \text{ odd},\ z \text{ odd}\} \quad \text{(the dicyclic group of order 12)}.
\end{aligned}$$

Below each N_i.

- $N_1 \cong \mathbb{Z}_{12}$ has 3 composition series (part (a)): through $Z \subset C$, through $Z \subset C_4$, and through $P \subset C$.
- N_2: a normal subgroup of order 4 would be a normal Sylow 2-subgroup, which does not exist, so G_2 has order 6 and index 2 in N_2. The three subgroups of index 2 in N_2 are C, S, and T. Now $C \cong \mathbb{Z}_6$ has the two series $\{e\} \subset Z \subset C$ and $\{e\} \subset P \subset C$, while $S \cong T \cong S_3$ each have only $\{e\} \subset P \subset S$ and $\{e\} \subset P \subset T$ (since S_3 has no normal subgroup of order 2). That gives 4 series.
- N_3: its Sylow 2-subgroups are not normal and it has a unique subgroup of index 2, namely C, so there are 2 series (through $Z \subset C$ and through $P \subset C$).

Altogether G has $3 + 4 + 2 = 9$ composition series:

$$\begin{array}{lll} \{e\} \subset Z \subset C \subset N_1 \subset G, & \{e\} \subset Z \subset C_4 \subset N_1 \subset G, & \{e\} \subset P \subset C \subset N_1 \subset G, \\ \{e\} \subset Z \subset C \subset N_2 \subset G, & \{e\} \subset P \subset C \subset N_2 \subset G, & \{e\} \subset P \subset S \subset N_2 \subset G, \\ \{e\} \subset P \subset T \subset N_2 \subset G, & \{e\} \subset Z \subset C \subset N_3 \subset G, & \{e\} \subset P \subset C \subset N_3 \subset G. \end{array}$$

(f) S_4**.** The normal subgroups of S_4 are $\{e\}, V_4, A_4, S_4$, where $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$. In a composition series, G_{k-1} must be a normal subgroup with S_4/G_{k-1} simple, which forces $G_{k-1} = A_4$ (as $S_4/V_4 \cong S_3$ is not simple). Then G_{k-2} is normal in A_4 with simple quotient; the normal subgroups of A_4 are $\{e\}, V_4, A_4$, and $A_4/V_4 \cong \mathbb{Z}_3$, so $G_{k-2} = V_4$. Finally G_{k-3} is a subgroup of order 2 of V_4 (automatically normal in the abelian group V_4), and there are three of them. So there are exactly 3 composition series:

$$\begin{aligned} &\{e\} \subset \langle (12)(34) \rangle \subset V_4 \subset A_4 \subset S_4, \\ &\{e\} \subset \langle (13)(24) \rangle \subset V_4 \subset A_4 \subset S_4, \\ &\{e\} \subset \langle (14)(23) \rangle \subset V_4 \subset A_4 \subset S_4, \end{aligned}$$

each with factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_2$.

(g) S_n**,** $n \geq 5$**.** There is exactly one composition series:

$$\{e\} \subset A_n \subset S_n, \qquad \text{with factors } A_n \text{ and } \mathbb{Z}_2.$$

A_n is simple for $n \geq 5$. We claim the only normal subgroups of S_n are $\{e\}$, A_n, S_n. Let $N \trianglelefteq S_n$ be non-trivial. Then $N \cap A_n \trianglelefteq A_n$, so it is $\{e\}$ or A_n. If $N \cap A_n = \{e\}$, then N embeds in $S_n/A_n \cong \mathbb{Z}_2$, so $N = \{e, x\}$ with x central; but $Z(S_n)$ is trivial for $n \geq 3$, a contradiction. So $N \supseteq A_n$, and $N = A_n$ or S_n.

Now in a composition series, G_{k-1} is a normal subgroup of S_n with S_n/G_{k-1} simple. $G_{k-1} = \{e\}$ is impossible since S_n is not simple, so $G_{k-1} = A_n$ (and $S_n/A_n \cong \mathbb{Z}_2$ is simple). Since A_n is simple, the next term is $\{e\}$.

(h) $\mathbb{Q}$ (under addition). $\mathbb{Q}$ has no composition series. Suppose it had one, $\{0\} = G_0 \subset \cdots \subset G_k = \mathbb{Q}$; since $\mathbb{Q} \neq \{0\}$, $k \geq 1$ and $H = G_{k-1}$ is a proper subgroup with $\mathbb{Q}/H$ simple. Since $\mathbb{Q}/H$ is abelian and simple, it is cyclic of prime order p, so $p(x + H) = H$ for every $x \in \mathbb{Q}$, that is, $px \in H$ for all $x \in \mathbb{Q}$. But every $y \in \mathbb{Q}$ can be written $y = p \cdot (y/p)$, so $y \in H$. Hence $H = \mathbb{Q}$, a contradiction.

Exercise 5

Show that the infinite direct product $G = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \cdots$ is not finitely generated.

Solution

Suppose G is generated by $g_1, \ldots, g_m$. Since G is abelian and $2g_i = 0$ for every i, every element of G can be written as $\varepsilon_1 g_1 + \cdots + \varepsilon_m g_m$ with each $\varepsilon_i \in \{0, 1\}$. So $|G| \leq 2^m$ is finite. But G is infinite (it contains the distinct elements $e_n = (0, \ldots, 0, 1, 0, \ldots)$, with the 1 in position n, for each $n \geq 1$). This contradiction shows G is not finitely generated. (The same argument applies to the infinite direct *sum*.) □

Exercise 6

Let G be an abelian group of order m. If n divides m, prove that G has a subgroup of order n.

Solution

By the Fundamental Theorem, $G \cong \bigoplus_p G_p$ over the primes $p \mid m$, where $|G_p| = p^{a_p}$ and $G_p \cong \mathbb{Z}_{p^{e_1}} \times \cdots \times \mathbb{Z}_{p^{e_r}}$ with $e_1 + \cdots + e_r = a_p$. Write $n = \prod p^{b_p}$; since $n \mid m$, $0 \leq b_p \leq a_p$.

For each p, choose integers $0 \leq f_i \leq e_i$ with $f_1 + \cdots + f_r = b_p$ (possible since $0 \leq b_p \leq e_1 + \cdots + e_r$). In $\mathbb{Z}_{p^{e_i}}$, the subgroup $\langle p^{e_i - f_i} \rangle$ has order p^{f_i}. So $H_p = \prod_i \langle p^{e_i - f_i} \rangle \leq G_p$ has order $p^{f_1 + \cdots + f_r} = p^{b_p}$. Then $H = \bigoplus_p H_p$ is a subgroup of G of order $\prod p^{b_p} = n$. □

Exercise 7

A group G is a torsion group if every element of G has finite order. Prove that a finitely generated abelian torsion group must be finite.

Solution

Let $G = \langle g_1, \ldots, g_m \rangle$ be abelian, with g_i of order $n_i < \infty$. Since G is abelian, every element can be written $g_1^{a_1} \cdots g_m^{a_m}$, and reducing exponents mod n_i we may take $0 \leq a_i < n_i$. Thus $|G| \leq n_1 n_2 \cdots n_m < \infty$. □

Exercise 8

Let G, H, and K be finitely generated abelian groups. Show that if $G \times H \cong G \times K$, then $H \cong K$. Give a counterexample to show that this cannot be true in general.

Solution

By the Fundamental Theorem, write each finitely generated abelian group X as $\mathbb{Z}^{r(X)} \times T(X)$, with $T(X)$ finite and determined by its multiset $E(X)$ of prime-power orders of cyclic factors (the elementary divisors); the number $r(X)$ and the multiset $E(X)$ are invariants of the isomorphism class of X. For a direct product,

$$r(G \times H) = r(G) + r(H), \qquad E(G \times H) = E(G) \uplus E(H) \text{ (multiset union).}$$

If $G \times H \cong G \times K$, these invariants agree, so $r(G) + r(H) = r(G) + r(K)$ and $E(G) \uplus E(H) = E(G) \uplus E(K)$. Cancelling gives $r(H) = r(K)$ and $E(H) = E(K)$, hence $H \cong K$. □

Counterexample without finite generation. Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \cdots$ (infinite product), $H = \mathbb{Z}_2$, $K = \{0\}$. The map

$$G \times \mathbb{Z}_2 \to G, \qquad \big((x_1, x_2, \dots), y\big) \mapsto (y, x_1, x_2, \dots)$$

is an isomorphism, so $G \times H \cong G \cong G \times K$, but $H = \mathbb{Z}_2 \ncong \{0\} = K$. (Similarly, with $G = \mathbb{Z} \oplus \mathbb{Z} \oplus \cdots$, $H = \mathbb{Z}$, $K = \{0\}$.)

Exercise 10

If G has a composition (principal) series and if N is a proper normal subgroup of G, show there exists a composition (principal) series containing N.

Solution

Let $\pi : G \to G/N$ be the canonical map. By Exercise 17 below, N and G/N both have composition series:

$$\{e\} = N_0 \triangleleft N_1 \triangleleft \cdots \triangleleft N_m = N, \qquad \{N\} = \bar{G}_0 \triangleleft \bar{G}_1 \triangleleft \cdots \triangleleft \bar{G}_k = G/N.$$

Put $H_i = \pi^{-1}(\bar{G}_i)$. By Fact A, $N = H_0 \triangleleft H_1 \triangleleft \cdots \triangleleft H_k = G$ and $H_{i+1}/H_i \cong \bar{G}_{i+1}/\bar{G}_i$ is simple. Therefore

$$\{e\} = N_0 \triangleleft \cdots \triangleleft N_m = N = H_0 \triangleleft H_1 \triangleleft \cdots \triangleleft H_k = G$$

is a composition series of G, and it contains N. □

Principal series. The same argument works for a principal series (a normal series with simple factors). The series $N \cap G_i$ used in Exercise 17 consists of subgroups normal in G when each G_i is, and the subgroups $H_i = \pi^{-1}(\bar{G}_i)$ are normal in G when each $\bar{G}_i$ is normal in G/N; the factor-group arguments are unchanged.

Exercise 11

Prove or disprove: Let N be a normal subgroup of G. If N and G/N have composition series, then G must also have a composition series.

Solution

This is true. Take composition series $\{e\} = N_0 \triangleleft \cdots \triangleleft N_m = N$ and $\{N\} = \bar{G}_0 \triangleleft \cdots \triangleleft \bar{G}_k = G/N$, and let $H_i = \pi^{-1}(\bar{G}_i)$. By Fact A, $H_i \triangleleft H_{i+1}$ with $H_{i+1}/H_i \cong \bar{G}_{i+1}/\bar{G}_i$ simple, and $H_0 = N$, $H_k = G$. Then

$$\{e\} = N_0 \triangleleft \cdots \triangleleft N_m = N = H_0 \triangleleft H_1 \triangleleft \cdots \triangleleft H_k = G$$

is a composition series of G. □

(Together with Exercise 17, this shows: G has a composition series if and only if N and G/N do.)

Exercise 17

Suppose that G has a composition series. If N is a normal subgroup of G, show that N and G/N also have composition series.

Solution

Let $\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$ be a composition series.

N has a composition series. Put $N_i = N \cap G_i$, so $\{e\} = N_0 \subseteq N_1 \subseteq \cdots \subseteq N_n = N$. Each N_i is normal in N_{i+1}: for $x \in N_{i+1}$ and $y \in N_i$, $xyx^{-1} \in N$ (as $x, y \in N$) and $xyx^{-1} \in G_i$ (as $x \in G_{i+1}$, $y \in G_i$, $G_i \triangleleft G_{i+1}$). Define

$$f : N_{i+1} \to G_{i+1}/G_i, \qquad f(x) = xG_i.$$

Its kernel is $N_{i+1} \cap G_i = N_i$, so $N_{i+1}/N_i \cong f(N_{i+1})$. Moreover $f(N_{i+1})$ is normal in G_{i+1}/G_i: for $g \in G_{i+1}$ and $x \in N_{i+1}$, $gxg^{-1} \in N \cap G_{i+1} = N_{i+1}$ because $N \triangleleft G$, and $(gG_i)f(x)(gG_i)^{-1} = f(gxg^{-1})$. Since G_{i+1}/G_i is simple, $f(N_{i+1})$ is trivial or everything. Thus each factor N_{i+1}/N_i is either trivial or isomorphic to the simple group G_{i+1}/G_i. Deleting repeated terms ($N_i = N_{i+1}$) from the chain gives a composition series of N.

G/N has a composition series. Let $\pi : G \to G/N$ and $\bar{G}_i = \pi(G_i)$, so $\{N\} = \bar{G}_0 \subseteq \cdots \subseteq \bar{G}_n = G/N$. Each $\bar{G}_i$ is normal in $\bar{G}_{i+1}$: for $x \in G_{i+1}$ and $y \in G_i$, $\pi(x)\pi(y)\pi(x)^{-1} = \pi(xyx^{-1}) \in \bar{G}_i$. The map

$$G_{i+1} \to \bar{G}_{i+1}/\bar{G}_i, \qquad x \mapsto \pi(x)\bar{G}_i$$

is a surjective homomorphism whose kernel contains G_i, so it induces a surjective homomorphism $G_{i+1}/G_i \to \bar{G}_{i+1}/\bar{G}_i$. Its kernel is normal in the simple group G_{i+1}/G_i, so it is trivial or everything. Hence $\bar{G}_{i+1}/\bar{G}_i$ is isomorphic to G_{i+1}/G_i or is trivial. Deleting repeated terms gives a composition series of G/N. □

Exercise 18

Let G be a cyclic p-group with subgroups H and K. Prove that either H is contained in K or K is contained in H.

Solution

Let $G = \langle g \rangle$ have order p^n. Every subgroup L of a cyclic group is cyclic, generated by g^m where m is the smallest positive integer with $g^m \in L$ (or $L = \{e\}$). Since $g^{p^n} = e \in L$, m divides p^n (by the division algorithm, m divides every exponent a with $g^a \in L$), so $m = p^i$ for some $0 \le i \le n$. Hence every subgroup of G has the form $\langle g^{p^i} \rangle$ for some $0 \le i \le n$ (with $i = n$ giving $\{e\}$).

Now let $H = \langle g^{p^i} \rangle$ and $K = \langle g^{p^j} \rangle$, and say $i \ge j$. Then

$$g^{p^i} = \left(g^{p^j}\right)^{p^{i-j}} \in K,$$

so $H \subseteq K$. If instead $i \le j$, the same argument gives $K \subseteq H$. □

Exercise 22 (Zassenhaus Lemma)

Let H and K be subgroups of a group G. Suppose also that H^* and K^* are normal subgroups of H and K respectively. Then

(a) $H^*(H \cap K^*)$ is a normal subgroup of $H^*(H \cap K)$.

(b) $K^*(H^* \cap K)$ is a normal subgroup of $K^*(H \cap K)$.

(c) $H^*(H \cap K)/H^*(H \cap K^*) \;\cong\; K^*(H \cap K)/K^*(H^* \cap K) \;\cong\; (H \cap K)/(H^* \cap K)(H \cap K^*)$.

Solution

Preliminaries. If $N \trianglelefteq A$ and $L \leq A$, then NL is a subgroup of A; if L is also normal in A, then NL is normal in A and $NL = LN$. Hence $H^*(H \cap K)$ and $H^*(H \cap K^*)$ are subgroups of H (since $H^* \trianglelefteq H$), and $K^*(H \cap K)$ and $K^*(H^* \cap K)$ are subgroups of K.

Let

$$D = (H^* \cap K)(H \cap K^*).$$

Claim: $D \trianglelefteq H \cap K$. If $x \in H \cap K$ and $y \in H^* \cap K$, then $xyx^{-1} \in H^*$ (as $x \in H$ and $H^* \trianglelefteq H$) and $xyx^{-1} \in K$ (as $x, y \in K$), so $H^* \cap K \trianglelefteq H \cap K$. In the same way, $H \cap K^* \trianglelefteq H \cap K$ (using $K^* \trianglelefteq K$). The product of two normal subgroups of $H \cap K$ is normal in $H \cap K$, which proves the claim. Note also that $D = (H \cap K^*)(H^* \cap K) = (K^* \cap H)(K \cap H^*)$, so D is symmetric in the roles of (H, H^*) and (K, K^*).

The map. Define

$$\varphi : H^*(H \cap K) \to (H \cap K)/D, \qquad \varphi(h^*x) = xD \quad (h^* \in H^*,\ x \in H \cap K).$$

Well defined. Suppose $h^*x = h_1^*x_1$ with $h^*, h_1^* \in H^*$ and $x, x_1 \in H \cap K$. Then $x_1x^{-1} = (h_1^*)^{-1}h^* \in H^* \cap (H \cap K) = H^* \cap K \subseteq D$. Since $D \trianglelefteq H \cap K$, $x^{-1}x_1 = x^{-1}(x_1x^{-1})x \in D$, so $xD = x_1D$.

Homomorphism. For $h^*, h_1^* \in H^*$ and $x, x_1 \in H \cap K$,

$$(h^*x)(h_1^*x_1) = h^*(xh_1^*x^{-1})\,xx_1, \qquad xh_1^*x^{-1} \in H^* \text{ (since } x \in H,\ H^* \trianglelefteq H),$$

so $\varphi\big((h^*x)(h_1^*x_1)\big) = xx_1D = (xD)(x_1D)$.

Onto. $\varphi(x) = xD$ for every $x \in H \cap K$.

Kernel. $h^*x \in \ker\varphi$ if and only if $x \in D$. So

$$\ker\varphi = \{h^*x : h^* \in H^*,\ x \in D\} = H^*D = H^*(H^* \cap K)(H \cap K^*) = H^*(H \cap K^*),$$

because $H^*(H^* \cap K) = H^*$.

(a) A kernel is normal, so $H^*(H \cap K^*) = \ker\varphi \trianglelefteq H^*(H \cap K)$.

(c), first isomorphism. By the First Isomorphism Theorem,

$$H^*(H \cap K)/H^*(H \cap K^*) \ \cong\ (H \cap K)/D = (H \cap K)/(H^* \cap K)(H \cap K^*).$$

(b) and (c), second isomorphism. Interchange the roles of (H, H^*) and (K, K^*) in the argument above. Since D is unchanged by this interchange, we obtain $K^*(H^* \cap K) \trianglelefteq K^*(H \cap K)$, which is (b), and

$$K^*(H \cap K)/K^*(H^* \cap K) \ \cong\ (H \cap K)/D.$$

Combining the two isomorphisms gives (c). □

Exercise 23 (Schreier's Theorem)

Use the Zassenhaus Lemma to prove that two subnormal (normal) series of a group G have isomorphic refinements.

Solution

Let

$$G = H_0 \supseteq H_1 \supseteq \cdots \supseteq H_n = \{e\}, \qquad G = K_0 \supseteq K_1 \supseteq \cdots \supseteq K_m = \{e\}$$

be two subnormal series, with $H_{i+1} \trianglelefteq H_i$ and $K_{j+1} \trianglelefteq K_j$. Define, for $0 \le i < n$ and $0 \le j \le m$,

$$H_{i,j} = H_{i+1}(H_i \cap K_j),$$

and, for $0 \le j < m$ and $0 \le i \le n$,

$$K_{j,i} = K_{j+1}(K_j \cap H_i).$$

(These are subgroups, by the preliminaries in Exercise 22.)

Refinement of the H-series. We have

$$H_{i,0} = H_{i+1}(H_i \cap G) = H_{i+1}H_i = H_i, \qquad H_{i,m} = H_{i+1}(H_i \cap \{e\}) = H_{i+1},$$

and $H_{i,j+1} = H_{i+1}(H_i \cap K_{j+1}) \subseteq H_{i+1}(H_i \cap K_j) = H_{i,j}$. Apply Exercise 22(a) with $H = H_i$, $H^* = H_{i+1}$, $K = K_j$, $K^* = K_{j+1}$: it says

$$H_{i+1}(H_i \cap K_{j+1}) \ \trianglelefteq \ H_{i+1}(H_i \cap K_j), \quad \text{that is,} \quad H_{i,j+1} \trianglelefteq H_{i,j}.$$

So for each i,

$$H_i = H_{i,0} \supseteq H_{i,1} \supseteq \cdots \supseteq H_{i,m} = H_{i+1}$$

is a subnormal chain, and putting these chains together for $i = 0, \ldots, n-1$ gives a subnormal series

$$G = H_{0,0} \supseteq \cdots \supseteq H_{0,m} = H_{1,0} \supseteq \cdots \supseteq H_{1,m} = H_{2,0} \supseteq \cdots \supseteq H_{n-1,m} = \{e\}$$

that contains every H_i, that is, a refinement $\mathcal{H}'$ of the H-series, with nm factor groups $H_{i,j}/H_{i,j+1}$.

Refinement of the K-series. Symmetrically, $K_{j,0} = K_j$, $K_{j,n} = K_{j+1}$, $K_{j,i+1} \subseteq K_{j,i}$, and Exercise 22(b) (same assignment of H, H^*, K, K^*) says

$$K_{j+1}(H_{i+1} \cap K_j) \trianglelefteq K_{j+1}(H_i \cap K_j), \quad \text{that is,} \quad K_{j,i+1} \trianglelefteq K_{j,i}.$$

This gives a refinement $\mathcal{K}'$ of the K-series with mn factor groups $K_{j,i}/K_{j,i+1}$.

The refinements are isomorphic. Apply Exercise 22(c) with the same assignment $H = H_i$, $H^* = H_{i+1}$, $K = K_j$, $K^* = K_{j+1}$. Its left and middle terms are

$$H^*(H \cap K)/H^*(H \cap K^*) = H_{i,j}/H_{i,j+1}, \qquad K^*(H \cap K)/K^*(H^* \cap K) = K_{j,i}/K_{j,i+1},$$

so

$$H_{i,j}/H_{i,j+1} \ \cong \ K_{j,i}/K_{j,i+1} \qquad \text{for all } 0 \le i < n, \ 0 \le j < m.$$

Thus $\mathcal{H}'$ and $\mathcal{K}'$ have the same number nm of factor groups, and the factor in position (i,j) of $\mathcal{H}'$ is isomorphic to the factor in position (j,i) of $\mathcal{K}'$. This is a bijection between the factor groups of the two series matching isomorphic groups, so $\mathcal{H}'$ and $\mathcal{K}'$ are isomorphic refinements. (Repeated terms, which correspond to trivial factors, may be deleted from both; trivial factors correspond to trivial factors, so the remaining series are still isomorphic.)

Normal series. If every H_i and K_j is normal in G, then $H_i \cap K_j$ is normal in G, and a product of normal subgroups of G is normal in G. So every $H_{i,j}$ and $K_{j,i}$ is normal in G, and the refinements above are normal series. $\square$

Exercise 24 (Jordan–Hölder Theorem)

Use Schreier's Theorem to prove the Jordan–Hölder Theorem: any two composition series of a group G are isomorphic.

Solution

Let

$$G = H_0 \supset H_1 \supset \cdots \supset H_n = \{e\}, \qquad G = K_0 \supset K_1 \supset \cdots \supset K_m = \{e\}$$

be two composition series: each $H_{i+1} \trianglelefteq H_i$ and $K_{j+1} \trianglelefteq K_j$, and every factor group is simple (in particular non-trivial).

Step 1: a refinement of a composition series adds only repeated terms. Fix i and let

$$H_i = L_0 \supseteq L_1 \supseteq \cdots \supseteq L_r = H_{i+1}$$

be a subnormal chain, $L_{t+1} \trianglelefteq L_t$, inserted between H_i and H_{i+1}. We show by induction on t that each L_t equals H_i or H_{i+1}. This is true for $t = 0$. Suppose $L_t \in \{H_i, H_{i+1}\}$.

- If $L_t = H_{i+1}$, then $H_{i+1} = L_r \subseteq L_{t+1} \subseteq L_t = H_{i+1}$, so $L_{t+1} = H_{i+1}$.
- If $L_t = H_i$, then $L_{t+1} \trianglelefteq H_i$ and $H_{i+1} \subseteq L_{t+1}$, so L_{t+1}/H_{i+1} is a normal subgroup of the simple group H_i/H_{i+1}. Hence L_{t+1}/H_{i+1} is trivial or all of H_i/H_{i+1}, that is, $L_{t+1} \in \{H_{i+1}, H_i\}$.

So after deleting repeated terms, any refinement of a composition series is the series itself. Equivalently, every refinement has the original factors together with trivial factors.

Step 2. By Schreier's Theorem (Exercise 23), the two composition series have refinements $\mathcal{H}'$ and $\mathcal{K}'$ that are isomorphic: there is a bijection between their factor groups matching isomorphic groups. By Step 1, the non-trivial factors of $\mathcal{H}'$ are exactly the factors H_i/H_{i+1} $(0 \le i < n)$, each simple, and the non-trivial factors of $\mathcal{K}'$ are exactly the factors K_j/K_{j+1} $(0 \le j < m)$. The bijection sends trivial factors to trivial factors (a group isomorphic to a trivial group is trivial), so it restricts to a bijection between the non-trivial factors. Hence $n = m$, and after reordering, $H_i/H_{i+1} \cong K_{\sigma(i)}/K_{\sigma(i)+1}$ for some permutation σ of $\{0, \ldots, n-1\}$. That is, the two composition series are isomorphic. □